

Compositional Correctness via Directed Containers: A Complete Technical Dossier

MacBeth
(with N. Ghani, R. Langer)
Kodamai

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Abstract

This dossier collects the container / directed-container research programme into a single document, organised around one thesis: the correctness of a composite computational system is governed by a single, checkable equation,

$$\text{a composite is correct} \iff (\mathbf{L}) \wedge [\omega] = 0 \in H^2.$$

The document has four pillars — Theory, Applications, Formalisation, Impact — and an honest ledger of what is proved, what is machine-verified, and what remains open. This is a *skeleton*: the front matter is written in full, and each downstream section names the source file(s) from which its body will be assembled.

Contents

How to read this document	3
1 Overview: One Thesis	4
1.1 The organizing equation	4
1.2 Why this is a foundation and not a slogan	4
1.3 The four movements	5
2 The Honest Ledger	6
I Theory	7
3 Preliminaries: the machinery	8
3.1 Representable functors and the Yoneda lemma	8
3.2 Density, the extension functor, and $\text{Fam}(\mathbf{Setop})$	9
3.3 Directed containers and the composition operator $\triangleleft$	10
3.4 Day convolution and the four monoidal structures	11
3.5 Cohomological conventions	12
4 The equivalence chain	13
5 Zappa–Szép products and distributive laws as the composition tool	14
6 The correctness criterion $(L) \wedge [\omega] = 0 \in H^2$	15
7 The H^2-cluster of obstruction theories	16
7.1 Three homes on one object	16
7.2 The refuted unifier, and the honest umbrella	17
7.3 The crown: the cluster is filtered, not a product	18
8 Change-of-base, $\text{Fam}(\mathcal{C}^{\text{op}})$ generality, and the $\{\text{Id}\}$ classification	19
9 The set-theoretic frontier: Vec-admissibility and the Gap-S arc	20
9.1 From admissibility to a module	20
9.2 The first count: false under CH	22
9.3 The sharp invariant: $\mu_S(M) = \mathfrak{d}$	22
9.4 The sharpened refutation, and why it matters	23
9.5 The residual, honestly	24

II	Applications	26
10	GA / island-model composition	27
11	Blockchain: re-entrancy as a failed distributive law	28
12	Tax computation (Catala)	29
13	Agent orchestration: the harness is a worker	30
14	Security audits as failed distributive laws	31
15	Supply chains	32
III	Formalisation	33
16	Milestones M1–M6: status	34
17	The sorry-free Lean corpus	35
18	H^2 -engines and the free/cofree comonad in Lean	36
19	Change-of-base adjunction and collapse theorems in Lean	37
IV	Impact	38
20	Where compositional correctness has economic consequences	39
21	The AI + Lean pipeline: honesty and trust-grade methodology	40
21.1	The problem this solves	40
21.2	The trust-grade ladder	40
21.3	The registry	41
21.4	The Lean statement self-audit	41
21.5	Forcing functions: honesty made structural	42
21.6	“Correction is the theorem”	42
22	External-publication status	44

How to read this document

This is a document being assembled in stages. It compiles at every stage, and several chapters are already written in full: the front matter (this chapter, the Overview, and the Honest Ledger), the preliminaries (Chapter 3), the H^2 -cluster (Chapter 7), the set-theoretic frontier (Chapter 9), and the AI + Lean pipeline (Chapter 21). These are finished prose meant to be read.

The remaining chapters are *scaffolded*: each opens with a short genuine lead-in and then carries a machine-readable assembly directive of the form

`% ASSEMBLE FROM: <path> [grade] [ASSEMBLE|STUB|GAP]`

naming the source paper(s) whose bodies will be pulled in during a later assembly session, together with the strongest current trust grade of the material and one of three assembly tags: ASSEMBLE (a written source exists and mainly needs merging), STUB (partial material exists; writing is required), or GAP (no artifact yet; must be drafted or framed as illustrative). A reader can therefore tell at a glance which chapters are load-bearing and which are still promissory.

The trust grades. Throughout, claims carry a trust grade drawn from the programme’s standard ladder, in increasing order of external verifiability:

speculative < computed < proved < peer-reviewed < published < lean-verified.

Here *computed* means checked on explicit small cases or by symbolic computation; *proved* means a paper proof we believe complete; *peer-reviewed* means refereed internally (currently: Rick, a fellow agent); *published* means externally published (currently: nothing — see the Honest Ledger); and *lean-verified* means machine-checked, sorry-free, in Lean 4/Mathlib. The ordering places *lean-verified* at the top because a sorry-free proof term is the highest available warrant — with the essential caveat, stated plainly in Part IV, that “verified” certifies the proof, not that the *statement* is the one you meant.

Chapter 1

Overview: One Thesis

1.1 The organizing equation

The claim that unifies this programme is that the composition of heterogeneous computational systems is governed by a single equation, and that the equation is *checkable*:

$$\boxed{\text{a composite is correct} \iff (\mathbf{L}) \wedge [\omega] = 0 \in H^2}$$

The two conjuncts are independent and have different characters.

(L) — **the local condition.** Each part of the system is a *directed container* — equivalently, a small category, equivalently a polynomial comonoid. Condition (L) says that each part composes coherently *with itself*: its identities and its associative composition are the counit and comultiplication of a comonoid structure. This is the content of the equivalence chain

$$\mathbf{Cont} \simeq \mathbf{DCont} \simeq (\text{polynomial comonoids}) \simeq \mathbf{Cat},$$

which four independent lines of work reach from four directions. (L) is a condition on the pieces, in isolation.

$[\omega] = 0$ — **the global condition.** Given local pieces, the question is whether they *weld* into a single coherent whole. The weld is a Zappa–Szép product, equivalently a distributive law between the pieces’ comonads/monads. Such a weld exists precisely when a degree-two cohomology class $[\omega] \in H^2$ vanishes. Where (L) is about each part alone, $[\omega] = 0$ is about their interaction: it is the obstruction to gluing.

Everything in the programme is one of three things: a *proof* that this equation is the right one and holds where claimed; an *application* in which a real system’s failure mode is exactly $[\omega] \neq 0$; or a *formalisation* that makes the check machine-executable.

1.2 Why this is a foundation and not a slogan

The equation is not imposed; it is discovered at the meeting point of three independent research programmes. Container theory (Nottingham: Abbott–Altenkirch–Ghani), directed containers (Tallinn: Ahman–Uustalu), and the **Poly** programme (Topos Institute: Spivak) developed substantially the same mathematics in three dialects, and all three converge on the equivalence chain that furnishes

(L). Convergence is the evidence: when three groups arrive at the same object from three directions, the object is real, not an artifact of notation.

What the present programme adds is the *composition* question none of the three answered head-on: when do two of these objects compose, and what obstructs it? The answer is the second conjunct. The pairwise Zappa–Szép criterion decomposes composability into a local factor (L) and a global factor (G), and $(G) \Leftrightarrow [\omega] = 0 \in H^2$. That identification — composability as the vanishing of a concrete cohomological obstruction — is the thesis of the dossier.

1.3 The four movements

The document proceeds in four movements, ordered so that each pillar earns the next.

I. Theory — the equation and its domain of validity. The equivalence chain (the (L) side); Zappa–Szép products and distributive laws as *the* composition tool, giving the pairwise criterion $(L) \wedge (G)$ with $(G) \Leftrightarrow [\omega] = 0$; the H^2 -cluster of three genuinely distinct obstruction theories on $\mathbf{DCont} \simeq \mathbf{Cat}$; the change-of-base / $\mathbf{Fam}(\mathcal{C}^{\text{op}})$ generality question, with the sharp negative that the self-enriched full-and-faithful semantics collapses to $\{\text{Id}\}$; and, at the frontier, the set-theoretic boundary of the equation’s domain — the Vec-admissibility / Gap-S arc, where the composability boundary of a categorical semantics turns out to be a cardinal characteristic of the continuum. Theory ends on the live edge, not a solved one.

II. Applications — where $[\omega] \neq 0$ costs money. The same mathematics across five domains: genetic-algorithm island-model composition, blockchain re-entrancy as a failed distributive law, security audits as failed distributive laws, agent orchestration (the harness as a worker), and tax computation. Real systems fail exactly when the weld obstruction is nonzero.

III. Formalisation — the check is executable. The Lean milestones and the sorry-free corpus: what is machine-verified versus proved on paper, stated honestly. This is what distinguishes the grant from hand-waving.

IV. Impact — and a note on honesty. Where compositional correctness has economic consequences, and the AI + Lean-pipeline honesty methodology: the discipline of separating “verified” from “correctly stated” and “reusable,” which is the programme’s real differentiator.

Chapter 2

The Honest Ledger

This chapter is deliberately near the front, not buried in an appendix. A dossier that claims a unifying equation must be equally clear about where the equation is unproved, where the applications lack artifacts, and what has and has not been externally validated. Four items:

1. **The Gap-S / Vec-admissibility wall is OPEN.** Whether **Vec** is $\triangleleft$ -admissible reduces to a projectivity question over the full linear ring $S = \text{End}_k(E)$, and the module M in question has generation number $\mu_S(M) = \mathfrak{d}$, the dominating number. We have proved the conjecture (Conj. 6.2) is *false* under CH, under MA, and in the Cohen model, via cardinal counting (M projective $\Rightarrow 2^{\mathfrak{d}} = 2^{\aleph_0}$, so any model with $2^{\aleph_0} < 2^{\mathfrak{d}}$ refutes it). But the *full ZFC status is unproved*: the verdict currently leans *independent*, and the residual case is the random-real model ($\mathfrak{d} = \aleph_1 < \mathfrak{c}$). We do not yet know whether M is projective there. This is the programme’s deepest open problem, and it is open.
2. **Tax computation has NO formal artifact.** The Catala / tax-computation application (§II.3) is currently a design sketch — ontology DAG, scope composition, comonadic event-sourced projections — with no proof, no Lean, and no running code. It is framed throughout as illustrative-only.
3. **Equivalence-chain Lean milestones M5–M6 are unproved.** Milestones M1–M4 of the equivalence chain are lean-verified; M5 and M6 are not formalised. The chain is complete on paper but not machine-checked end to end.
4. **Nothing is externally published.** Under the ACM authorship rule an AI agent cannot be sole author, and the external submission queue is held by Robin. The strongest current grade anywhere in this dossier is *peer-reviewed*, meaning refereed internally by Rick (a fellow agent) — e.g. the holonomy-composition / Zappa–Szép bridge and the m -containers result. No claim here is *published* in the external sense.

A fifth, smaller honesty note carried from the Theory pillar: the conjectured *protomodular unifier* of the three H^2 obstruction theories was *refuted*. The three theories are a genuinely distinct structured family, not one theorem wearing three hats.

Part I

Theory

Chapter 3

Preliminaries: the machinery

This chapter fixes notation and recalls the standard tools the rest of the dossier leans on, in one place, so that the later chapters can move without pausing to reintroduce them. *Almost everything here is standard* and is collected for self-containedness: the Yoneda lemma is Yoneda’s and Mac Lane’s; density and the coend/Kan calculus follow Mac Lane [3], Riehl [4], and Loregian [2]; Day convolution is Day’s [1]. The only programme-internal content is the *container reading* of each tool — the translation into shapes and positions — which is *proved* and is what makes the later theorems computable. Two facts of Ghani’s are foregrounded because the composition machinery uses them as black boxes: that the container-extension functor *is* a left Kan extension along the representable embedding (§3.2), and that the closed structure of a Day category is *forced on representables and extended by density* (§3.4).

The reader who wants a map: §3.1–3.2 (representables, Yoneda, density, the extension functor) are used everywhere the (L) side appears — the equivalence chain of Chapter 4, and, in an essential and less obvious way, the module reduction of the set-theoretic frontier (Chapter 9), where $\text{Nat}(F, -) \cong \text{Hom}_S(F(E), -)$ is Yoneda in disguise. Section 3.3 fixes the directed-container and composition ($\triangleleft$) notation carried by the Zappa–Szép chapters. Section 3.4 (Day convolution) supports the change-of-base chapter (Chapter 8). Section 3.5 fixes the cohomological conventions — H^2 , the class $[\omega]$, the Lyndon–Hochschild–Serre transgression — on which the correctness criterion and the H^2 -cluster of Chapter 7 stand.

Conventions. **Set** is the category of sets and functions. For a locally small category $\mathcal{C}$ we write $\mathcal{C}(a, b)$ for its hom-set; $\text{ob } \mathcal{C}$ for its objects. Functors are covariant unless a superscript op says otherwise. A *presheaf* on $\mathcal{C}$ is a functor $\text{Cop} \rightarrow \mathbf{Set}$; a *copresheaf* is a functor $\mathcal{C} \rightarrow \mathbf{Set}$. We write $\cong$ for canonical (natural) isomorphism and are careful to say *which* isomorphism when it matters.

3.1 Representable functors and the Yoneda lemma

Every construction in the dossier eventually reduces to a hom-set. The representable functors are the functors that *are* a hom-set with one argument held fixed, and they generate everything else.

Definition 3.1 (Representable copresheaf). Fix a set A . The *covariant representable* at A is

$$y^A := \mathbf{Set}(A, -) : \mathbf{Set} \longrightarrow \mathbf{Set}, \quad y^A(X) = \mathbf{Set}(A, X) = X^A,$$

sending $f: X \rightarrow Y$ to post-composition $y^A(f) = f \circ (-)$. A copresheaf $F: \mathbf{Set} \rightarrow \mathbf{Set}$ is *representable* if $F \cong y^A$ for some A , the *representing object*.

Remark 3.2 (The container reading). A *container* is a pair (S, P) of a set S of *shapes* and a family $P: S \rightarrow \mathbf{Set}$ of *position sets*, with *extension*

$$\llbracket S, P \rrbracket : \mathbf{Set} \rightarrow \mathbf{Set}, \quad \llbracket S, P \rrbracket(X) = \sum_{s \in S} \mathbf{Set}(P(s), X).$$

The *representable container* has a single shape and A positions, $(S, P) = (1, A)$, and $\llbracket 1, A \rrbracket(X) = \mathbf{Set}(A, X) = y^A(X)$. So representable copresheaves are exactly the extensions of single-shape containers, and the coproduct in the general formula is a coproduct of *representables* indexed by shapes. This is the whole story, phrased below first as a Kan extension (§3.2) and then as a Day convolution (§3.4). The same extension formula, over a base $\mathcal{C}$ other than $\mathbf{Set}$, is the object of study of Chapter 9.

Definition 3.1 is uniform in A : a function $a: A' \rightarrow A$ induces, by *pre-composition*, a natural transformation $y^A \Rightarrow y^{A'}$. Thus $A \mapsto y^A$ is a functor $y^{(-)}: \mathbf{Setop} \rightarrow [\mathbf{Set}, \mathbf{Set}]$, the *Yoneda embedding*. It is genuinely an embedding, as the Yoneda lemma shows.

Theorem 3.3 (Yoneda lemma; standard). *For any functor $F: \mathbf{Set} \rightarrow \mathbf{Set}$ and any set A there is a bijection*

$$\Theta_{F,A} : \mathbf{Nat}(y^A, F) \xrightarrow{\cong} F(A), \quad \Theta(\alpha) = \alpha_A(\mathrm{id}_A),$$

natural in F and A .

Proof. The inverse sends $x \in F(A)$ to $\Psi(x)$ with $\Psi(x)_X(g) = F(g)(x)$ for $g: A \rightarrow X$; naturality of $\Psi(x)$ is functoriality, $F(h)(F(g)(x)) = F(h \circ g)(x)$. Then $\Theta(\Psi(x)) = F(\mathrm{id}_A)(x) = x$. Conversely, for $\alpha: y^A \Rightarrow F$ and $x = \alpha_A(\mathrm{id}_A)$, naturality of α at $g: A \rightarrow X$ chased on id_A gives $\alpha_X(g) = F(g)(x) = \Psi(x)_X(g)$, so $\Psi(x) = \alpha$. Naturality in F and A is the same chase. $\square$

Corollary 3.4 (Full faithfulness). $\mathbf{Nat}(y^A, y^B) \cong \mathbf{Set}(B, A)$, *naturally; the Yoneda embedding $y^{(-)}: \mathbf{Setop} \rightarrow [\mathbf{Set}, \mathbf{Set}]$ is fully faithful. Natural transformations between representables are exactly functions, in the opposite direction.*

Corollary 3.4 is the reason the container-morphism convention (Notation 3.6) has position maps running *backwards*: a map of representables is a backwards map of representing objects.

3.2 Density, the extension functor, and $\mathbf{Fam}(\mathbf{Setop})$

The Yoneda lemma says representables see everything. Density says they *are* everything, once colimits are allowed.

Theorem 3.5 (Density / co-Yoneda; standard). *Every functor $F: \mathbf{Set} \rightarrow \mathbf{Set}$ is canonically a colimit of representables:*

$$F \cong \int^{A \in \mathbf{Set}} F(A) \bullet y^A,$$

where $F(A) \bullet y^A$ is the $F(A)$ -fold copower of y^A . Equivalently F is the colimit of the representables $y^A \xrightarrow{\Psi(x)} F$ over the category of elements (A, x) , $x \in F(A)$.

Reference. The density formula, dual to Yoneda; Riehl [4, §6.2], Loregian [2, Ch. 2]. Its universal property is Theorem 3.3 read backwards: a cocone from $F(A) \bullet y^A$ to G is a family $F(A) \rightarrow \mathbf{Nat}(y^A, G) \cong G(A)$ natural in A , i.e. a natural transformation $F \Rightarrow G$. $\square$

The representables are thus a *dense generator* of $[\mathbf{Set}, \mathbf{Set}]$: two constructions that agree on representables and preserve the relevant colimits agree everywhere. This uniformity is the workhorse of the whole (L) side.

Notation 3.6 (**Cont** and $\mathbf{Fam}(\mathbf{Setop})$). A morphism of containers $(S, P) \rightarrow (T, Q)$ is a pair (u, v) with $u: S \rightarrow T$ a map of shapes and v a family $v_s: Q(us) \rightarrow P(s)$ of position maps *going backwards*. This is exactly an S -indexed family of objects of $\mathbf{Setop}$ together with a map of index sets and, over it, morphisms in $\mathbf{Setop}$; hence

$$\mathbf{Cont} \cong \mathbf{Fam}(\mathbf{Setop}),$$

the *free coproduct completion* of $\mathbf{Setop}$ — the universal category with small coproducts containing $\mathbf{Setop}$. Under $\llbracket - \rrbracket$, a family $(P(s))_{s \in S}$ maps to $\sum_{s \in S} y^{P(s)}$.

Theorem 3.7 (Extension is a left Kan extension; Ghani’s observation, proved). *Let $\mathcal{A}$ be a category and $Y: \mathcal{A} \rightarrow \mathbf{Fam}(\mathcal{A})$ the singleton embedding $a \mapsto (a)$. For any $F: \mathcal{A} \rightarrow \mathcal{E}$ into a category with small coproducts, the coproduct-preserving extension $\widehat{F}((a_s)_{s \in S}) = \coprod_{s \in S} F(a_s)$ is the left Kan extension $\mathbf{Lan}_Y F$, and it preserves representables ($\widehat{F} \circ Y \cong F$). Specialising $\mathcal{A} = \mathbf{Setop}$, $F = y^{(-)}$, and $\mathcal{E} = [\mathbf{Set}, \mathbf{Set}]$, the container-extension functor is that Kan extension:*

$$\llbracket - \rrbracket = \mathbf{Lan}_Y y^{(-)}, \quad \llbracket S, P \rrbracket = \coprod_{s \in S} y^{P(s)}.$$

Proof. By the pointwise coend formula [3, Ch. X] and the description of morphisms out of a singleton in a free coproduct completion, $\mathbf{Fam}(\mathcal{A})(Ya, (a_s)_s) \cong \coprod_s \mathcal{A}(a, a_s)$, so

$$(\mathbf{Lan}_Y F)((a_s)_s) \cong \int^a \left(\coprod_s \mathcal{A}(a, a_s) \right) \bullet F(a) \cong \coprod_s \int^a \mathcal{A}(a, a_s) \bullet F(a) \cong \coprod_s F(a_s),$$

the inner coend being density (Theorem 3.5) at a_s . Preservation of representables is the one-term case $S = 1$. For $\mathcal{A} = \mathbf{Setop}$ and $F = y^{(-)}$ this reads $\llbracket S, P \rrbracket = \coprod_s y^{P(s)}$, whose value at X is $\sum_s \mathbf{Set}(P(s), X) = \llbracket S, P \rrbracket(X)$ (Remark 3.2). A size remark: although $\mathbf{Setop}$ is not small, part of the statement is really the universal property of the free coproduct completion, which holds verbatim, and the coproducts landed on are small since shape sets are. $\square$

Two consequences are used without reproof throughout Part I: (i) any structure on $\mathbf{Setop}$ transported to **Cont** need only be defined on singletons and extended by coproducts, and the result is automatically the Kan extension; (ii) to check a functor out of **Cont** agrees with $\llbracket - \rrbracket$ it suffices to check agreement on representables plus preservation of coproducts. “A container is a coproduct of representables” is not a slogan but a computational method.

3.3 Directed containers and the composition operator $\triangleleft$

The extension functors above are the raw material; the (L) side of the correctness criterion asks which of them compose *with themselves* coherently. That is the content of directed containers, whose full theory — including the equivalence chain — is Chapter 4. Here we fix only the notation the composition chapters need.

The category **Cont** carries a monoidal structure $\triangleleft$, *composition*, that on extensions is composition of endofunctors: $\llbracket p \triangleleft q \rrbracket \cong \llbracket p \rrbracket \circ \llbracket q \rrbracket$. Concretely, with $p = (S, P)$ and $q = (T, Q)$, a shape of $p \triangleleft q$ is a shape $s \in S$ together with a $P(s)$ -indexed family of shapes of q , and a position is a dependent pair — the “a shape, then a shape at each position” law that makes $\triangleleft$ the substitution tensor. The unit is the identity container $y^1 = (1, 1)$, whose extension is the identity functor.

Definition 3.8 (Directed container, informally). A *directed container* is a container p equipped with the structure of a $\triangleleft$ -comonoid: a counit $p \rightarrow y^1$ and a comultiplication $p \rightarrow p \triangleleft p$ satisfying the comonoid laws. Equivalently — and this is the equivalence chain — it is a small category, a polynomial comonoid in **Poly**, or a cofunctor domain, one object presented four ways. We write **DCont** for the category of directed containers.

Theorem 3.9 (The equivalence chain; assembled in Chapter 4). *There are equivalences of categories*

$$\mathbf{Cont} \supseteq \mathbf{DCont} \simeq (\text{polynomial comonoids in } \mathbf{Poly}) \simeq \mathbf{Cat},$$

where **Cat** is the category of small categories and (co)functors. The directed-container comonoid data are exactly a category's identities and composition.

This is the (L) condition: a system is composable-with-itself precisely when it is a directed container, equivalently a small category. The proof, and the four independent lines of work that converge on it, are the subject of Chapter 4; we state it here only so that later chapters may say “directed container” and “small category” interchangeably.

3.4 Day convolution and the four monoidal structures

Besides $\triangleleft$, the category **Cont** carries tensors transported from monoidal structures on the base by *Day convolution*. These are the setting of the change-of-base chapter (Chapter 8); we record only the degeneration that makes them coend-free.

Definition 3.10 (Day convolution; standard). For a small monoidal category $(\mathcal{C}, \otimes, I)$ the *Day convolution* of copresheaves F, G is

$$(F \otimes G)(c) = \int^{c_1, c_2} \mathcal{C}(c_1 \otimes c_2, c) \times F(c_1) \times G(c_2),$$

with unit y^I . Then $([\mathcal{C}, \mathbf{Set}], \otimes, y^I)$ is monoidal, cocontinuous in each variable, and $y^{(-)}$ is strong monoidal: $y^a \otimes y^b \cong y^{a \otimes b}$ [1].

Proposition 3.11 (Degenerate Day on containers; proved). *For any monoidal structure $\star$ on **Set**, Day convolution induces on $\mathbf{Cont} \cong \mathbf{Fam}(\mathbf{Setop})$ a monoidal structure $\odot_\star$ given without any coend by*

$$p \odot_\star q = (S_p \times S_q, (s, t) \mapsto p[s] \star q[t]),$$

where $p = (S_p, p[-])$, $q = (S_q, q[-])$. Shapes multiply; only the position law changes.

Proof. Write $p = \coprod_s y^{p[s]}$, $q = \coprod_t y^{q[t]}$ (Theorem 3.7). Day is cocontinuous in each variable, so $p \odot_\star q \cong \coprod_{s,t} (y^{p[s]} \odot_\star y^{q[t]}) \cong \coprod_{s,t} y^{p[s] \star q[t]}$ by strong monoidality on representables; read off the container by Theorem 3.7. The coend has been discharged on the generators. $\square$

Taking $\star \in \{+, \times\}$ gives two of the four tensors on **Cont** (the additive and Dirichlet convolutions); the substitution tensor $\triangleleft$ of §3.3 is a third, and it is *not* a Day convolution. The classification — which tensors arise, and which are closed — is Chapter 8. One fact from there is used as a black box: each Day tensor is closed, and its internal hom is *forced* on representables, $[y^a, H]_\otimes \cong H((-) \otimes a)$, then extended by density; for the Dirichlet tensor $[y^a, y^b]$ is the “shapes are morphisms” container with shape set $\mathbf{Cont}(y^a, y^b)$. Density *determines* this internal hom but does not hand it to every cocontinuous functor for free — preservation of a right adjoint is a separate, checkable fact — which is exactly the subtlety the $\{\text{Id}\}$ collapse of Chapter 8 turns on.

3.5 Cohomological conventions

The global condition $[\omega] = 0$ is a statement in low-degree cohomology. We fix conventions here; the theorems are Chapters 5–7.

For a group G and a G -module M we write $H^2(G; M)$ for the second group cohomology, classifying extensions $1 \rightarrow M \rightarrow E \rightarrow G \rightarrow 1$ up to equivalence; the class of an extension is its *Schreier class*. For a groupoid $\mathcal{G}$ we use the cohomology of its classifying space, which for a *connected* groupoid reduces to that of its vertex group π : $H^2(\mathcal{G}; M) \cong H^2(\pi; M)$. The class we call $[\omega]$ is the obstruction to a coherent weld of two systems — the Schreier class of the Zappa–Szép extension realising their composite; the criterion $(G) \iff [\omega] = 0$ is Chapter 6.

Two structural tools recur. For a normal extension $1 \rightarrow \mathcal{D} \rightarrow \pi \rightarrow \Gamma \rightarrow 1$ the *Lyndon–Hochschild–Serre spectral sequence* has $E_2^{p,q} = H^p(\Gamma; H^q(\mathcal{D}; M))$ converging to $H^{p+q}(\pi; M)$; its degree-two *transgression* d_2 is the differential that, in the programme’s identification, carries the holonomy ($d_2(\varphi) = \varphi_*[\omega]$). And the phrase “ $[\omega]$ is a differential, not a summand” means precisely that a base-edge class fails to survive to the abutment when $d_2 \neq 0$. These are the load-bearing facts behind the “filtration, not product” grammar of the H^2 -cluster (Chapter 7); we invoke them there by name.

Chapter 4

The equivalence chain

The local condition (L) rests on the identification of four objects that were developed independently: ordinary containers, directed containers, polynomial comonoids, and small categories. This chapter assembles the chain $\mathbf{Cont} \simeq \mathbf{DCont} \simeq (\text{poly comonoids}) \simeq \mathbf{Cat}$ and fixes the notation used throughout the dossier. It is the foundation on which every later composition statement stands: to compose systems coherently, each system must first *be* one of these objects.

[Assembly note: equivalence chain, M1–M4 lean-verified; assemble four dialects into one statement.]

Chapter 5

Zappa–Szép products and distributive laws as the composition tool

Given local pieces, composition is a *weld*, and the canonical weld of two categories/comonads is the Zappa–Szép product, equivalently a distributive law. This chapter develops the pairwise criterion: two directed containers compose iff the local condition (L) and a global condition (G) both hold. It is the machinery that makes “ $[\omega] = 0$ ” precise in the next chapter.

[Assembly note: pairwise-ZS proved; holonomy-composition / ZS bridge peer-reviewed (Rick).]

Chapter 6

The correctness criterion

$$(L) \wedge [\omega] = 0 \in H^2$$

Here the two threads join into the thesis equation. The global condition (G) of the pairwise criterion is identified with the vanishing of a degree-two cohomology class $[\omega] \in H^2$, so that a composite is correct exactly when the local condition holds and the weld obstruction vanishes. The chapter also records the “two ω -sites” result: the degree-1 (prunability / descent) and degree-2 (holonomy / Schreier) obstructions live at genuinely different sites and must not be conflated.

[Assembly note: two- ω -sites proved; the correctness equation is the chapter’s spine.]

Chapter 7

The H^2 -cluster of obstruction theories

The previous chapter distilled correctness to a single class, $[\omega] = 0 \in H^2$. That is one obstruction theory on the object $\mathbf{DCont} \simeq \mathbf{Cat}$. It is not the only one. Once the Zappa–Szép holonomy is in hand, two further degree-two obstruction theories present themselves on the *same* object — Ferri’s prunability obstruction and the direction-functor cohomology $H^2_{\mathbf{Cat}_B}$ — and the natural reflex is to hope they are three costumes of one theorem, three faces glued by a single universal H^2 . They are not. What this chapter records is the honest shape: a *structured family* of three genuinely distinct theories, with one proved separation between them, one refuted unifier (a valuable negative), and a single organising grammar that survives the refutation — the cluster is a *filtration*, not a product.

7.1 Three homes on one object

(a) **The multiplicative holonomy** $[\omega]$. The Zappa–Szép composition holonomy of Chapters 4–5 is a degree-two class $[\omega] \in H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D})$ over the orbit base $\mathrm{Sk}_{\mathcal{C}}$ (the skeleton of $\mathcal{C}$), with the criterion $(G) \iff [\omega] = 0$ (*proved*). It is a *multiplicative* invariant: it obstructs the existence of a coherent multiplicative transversal $\mathcal{C} \bowtie \mathcal{D}$. Two features fix its character. It vanishes for the *full loop bundle* and, more generally, on any *connected* groupoid, where the orbit base is coarse ($\simeq *$) and $H^{\geq 1}$ is forced to zero; it is nonzero only for a *proper* transversal $\mathcal{D} \subsetneq \pi$, where $\mathrm{Sk}_{\mathcal{C}} = B\Gamma$ carries the Schreier class of the extension $1 \rightarrow \mathcal{D} \rightarrow \pi \rightarrow \Gamma \rightarrow 1$ (nonzero already for $\mathbb{Z}/4$ over $\mathbb{Z}/2$) (*proved*). The degree-two computations underneath it are backed by a machine-checked engine: the cyclic and fibre H^2 modules are *lean-verified*, and the emergent-holonomy count is identified with Ext^0 together with a vanishing theorem killing the higher Ext-tower [17] (*proved; engine lean-verified*).

(b) **Ferri’s additive prunability**. Ferri’s prunability obstruction [18] asks whether the loop bundle G° of a quiver skew brace is an ideal — concretely, whether the loops over each object form an *additive* subgroup that the cross-action carries to loops. It is the *additive twin* of the multiplicative Schreier line of (a). Two facts pin its place in the cluster, and both are corrections of a tempting error.

First, it is *orthogonal* to $[\omega]$: on the full loop bundle the composition holonomy vanishes identically (Proposition A, *proved*) while prunability can still fail, so the two are separate obstructions — the existence of a connected, non-prunable witness that certifies the separation is *computed* [15]. Second, and sharper: prunability is *not* a degree-two class at all. In the untwisted core where pruning is even well posed, it is a *degree-one* datum — the condition that the loop subfunctor be $\dashv$ -flat, i.e. that G° be a *kernel* (a descent/parallel-transport condition), logically independent of the degree-two star-reassembly (H^2) class one degree above it [16] (*proved; the coset reformulation*).

proved, the Schreier-vs-descent crosstab computed). Reading Ferri’s non-prunability of $K_{2,3}$ as an “[ω] $\neq 0$ ” phenomenon is therefore a *category error (refuted)*: $K_{2,3}$ is connected, so [ω] = 0 there, and its obstruction lives one degree lower. *An ideal is a kernel; an extension is a reassembly.*

(c) **The direction-functor cohomology $H^2_{\mathbf{Cat}_B}$.** The third home is external machinery, not ours. Ambra, Duvieusart, and Montoli [19] equip the category $\mathbf{Cat}_B$ of internal categories with a cohomology relative to *Schreier points*: $\mathbf{Cat}_B$ is $\mathcal{S}$ -Mal’tsev there, and H^{n+1} is computed as π_0 of the fibres of a *direction functor*. For a connected groupoid this reduces to the vertex group, $H^2_{\mathbf{Cat}_B}(G; A) \cong H^2(\pi; M)$ (*proved; the external machinery deep-read*), placing $H^2_{\mathbf{Cat}_B}$ on the *additive*/vertex axis alongside prunability and complementary to the (base-axis) [ω]. The authors draw *no* link to containers or Zappa–Szép products, so this is not a scoop of our programme; it is reusable, independently developed infrastructure that the cluster can borrow.

7.2 The refuted unifier, and the honest umbrella

The three homes invite a single question: is there one protomodular / internal-semidirect-product H^2 that merges the multiplicative [ω] and the additive prunability into faces of a common class? The answer is *no*, and the negative is part of the result (*refuted, 2026-10-09*).

The structural reason is a clean divide. A connected groupoid splits canonically as $G \cong I(\mathrm{ob} G) \times B\pi$ — coarse base times vertex group — and the two obstruction theories are the cohomologies of the two complementary factors. But these factors sit on opposite sides of the *pointed* / *non-pointed* boundary. The vertex factor $B\pi$ is pointed, so the protomodular direction-functor $H^2(\pi; M)$ of (c) applies to it and carries the additive face. The base factor $\mathrm{Sk}_C = I(\mathrm{ob} G)$ is *non-pointed* — it has no zero object — so no protomodular H^2 sees it at all; [ω] is a nerve/simplicial class, vanishing exactly when the base is connected. A single semi-abelian H^2 cannot straddle the divide, because protomodularity is exactly the structure the non-pointed factor lacks.

What replaces the hoped-for single class is a *filtration*. The honest umbrella is a Künneth / product-nerve spectral sequence, whose abutment $H^2_{\mathrm{cat}}(G; M)$ is filtered rather than split. The natural target is

$$H^2_{\mathrm{cat}}(G; M) \cong H^2(\pi; M) \oplus H^2_{\mathrm{base}}(\mathrm{Sk}_C; M), \quad (7.1)$$

with the direction functor realising the vertex-group projection. This is registered as an open node and it is *speculative* — provable from the $H^2_{\mathbf{Cat}_B}$ reduction and the separation in the connected / field-coefficient case, where the filtration collapses, with genuine content only in the disconnected / proper-transversal case, where it does not. The refuted unifier and the speculative filtration are two readings of one fact: the coupling between the base and vertex faces is a *differential*, not a direct sum.

That last sentence is not a metaphor. On the normal Zappa–Szép slice the categorical cohomology $H^2_{\mathrm{cat}}(G) \cong H^2(\pi; M)$ is the abutment of the Lyndon–Hochschild–Serre spectral sequence of $1 \rightarrow \mathcal{D} \rightarrow \pi \rightarrow \Gamma \rightarrow 1$, whose two degree-two edges are exactly the two faces — the base edge $H^2(\Gamma; M^{\mathcal{D}})$ carrying [ω], the fibre edge $H^2(\mathcal{D}; M)^{\Gamma}$ carrying the additive reassembly class — and whose transgression *is* the holonomy, $d_2(\varphi) = \varphi_*[\omega]$ (*proved*) [13]. The naive direct sum (7.1) holds *if and only if* [ω] = 0; whenever [ω] $\neq 0$ the base edge does not survive to the abutment (it is a d_2 -image, not a summand), so the two faces are never simultaneously present as independent summands. The unifier fails for precisely the reason the differential is nonzero.

7.3 The crown: the cluster is filtered, not a product

The same grammar recurs one level up, and this is the connection that turns the cluster from a list into a structure (*proved, 2026-09-23*). Ferri's decomposition $G \cong (G/G^\odot) \bowtie G^\odot$ has an obstruction that is a *two-stage tower*, not a sum of two independent classes [14]:

- **Tier 2 (degree ≤ 1):** *is $G^\odot$ a kernel?* This is the $\rightarrow$ -descent condition of (b) — for $|L_\lambda| = 2$ a single degree-one cochain — and it vanishes exactly when the quotient $G/G^\odot$ *exists*.
- **Tier 1 (degree 2):** *given the quotient, does the loop extension split?* This is the class $[\omega] \in H^2$ of (a) — which is *only defined once Tier 2 vanishes*, and which is automatically zero on connected G .

The tiers are *filtered*: Tier 1 is undefined until Tier 2 vanishes, because the group $H^2(\text{Sk}_C; \mathcal{D})$ that houses $[\omega]$ is built from a quotient and a coefficient system that do not exist until $G^\odot$ is a kernel. A higher differential lives only on the kernel of the lower one — exactly the shape of the LHS transgression $d_2(\varphi) = \varphi_*[\omega]$ of the previous section, now read at the level of the decomposition itself. On connected G Tier 1 is free, so the decomposition exists iff G is prunable, recovering Ferri's Corollary 4.8 with its exact cohomological anatomy; $K_{2,3}$ is the canonical instance with Tier 1 = 0 and Tier 2 $\neq 0$ (*proved*).

This is the whole point of assembling the cluster. The three obstruction theories are not three costumes of one theorem, and they are not merged by one universal class; the unifier is refuted. What binds them is a single organising grammar — *filtration, not product* — appearing twice: once inside the abutment (the transgression coupling the two edges) and once above it (the two-tier tower coupling descent to splitting). A degree-one datum must vanish before the degree-two datum is even defined, and a nonzero degree-two class is a differential, not a summand. The honest deliverable of this chapter is that grammar, together with the proved separation and the honestly labelled negative that forced it.

[*Assembly note: separation proved (existence witness computed); prunability-is-degree-1-descent proved; $H^2_{\text{Cat}_B}$ reduction proved; protomodular unifier refuted; product decomposition (7.1) speculative; two-tier tower proved; H^2 -engines lean-verified.*]

Chapter 8

Change-of-base, $\mathbf{Fam}(\mathcal{C}^{\mathrm{op}})$ generality, and the $\{\mathrm{Id}\}$ classification

How far does the equation generalise? The natural setting is containers over a general base, i.e. $\mathbf{Fam}(\mathcal{C}^{\mathrm{op}})$, with change-of-base along functors and monads. This chapter assembles the positive generality results (m -containers, codensity/polynomiality, the four monoidal structures) and the sharp negative: the self-enriched full-and-faithful change-of-base container semantics collapses to $\{\mathrm{Id}\}$ — no nontrivial effect monad supports it. Knowing the boundary is itself the result.

[Assembly note: m -containers, change-of-base, fullness-unit-connectedness all proved; $\{\mathrm{Id}\}$ collapse is the sharp negative.]

Chapter 9

The set-theoretic frontier: Vec-admissibility and the Gap- S arc

This is the frontier of the Theory pillar, and it is open. Every obstruction so far in this document — the Zappa–Szép cocycle $[\omega] \in H^2$, the prunability descent class, the direction-functor cohomology — has been a concrete, *absolute* invariant: computed once, its verdict holds in every model of set theory. The admissibility of the linear base **Vec** is the first node of the programme that is not like this. Whether **Vec** is $\triangleleft$ -admissible turns out to be governed by the cardinal arithmetic of the continuum, and its exact truth value changes from one model of ZFC to another.

The result of this chapter is a single, sharp identity. The composability of the linear base reduces to the projectivity of one module M over the full linear ring $S = \text{End}_k(E)$, and the *minimal number of generators* of that module is exactly the dominating number:

$$\boxed{\mu_S(M) = \mathfrak{d}}.$$

A cardinal characteristic of the continuum is the load-bearing invariant of a functor-category admissibility question. From this identity the conjecture is *false* in every model with $\mathfrak{d} = \mathfrak{c}$ — in particular under CH, under Martin’s Axiom, and in the Cohen model — and the only surviving positive case is squeezed into the narrow window $2^{\aleph_0} = 2^{\mathfrak{d}}$, whose canonical witness is the random-real model. Whether M is projective there is the programme’s deepest open problem. The chapter ends the Theory pillar on that live edge, as it should.

Everything in §§9.1–9.3 is *proved* in ZFC; the residual of §9.5 is *open*. Nothing in this chapter is lean-verified, and nothing is externally published (Honest Ledger, items 1 and 4).

9.1 From admissibility to a module

Recall the denotation of a family $(S, P) \in \text{Fam}(\mathcal{C}^{\text{op}})$, the object that presents a container over the base $\mathcal{C}$:

$$\llbracket S, P \rrbracket X = \coprod_{s \in S} \mathcal{C}(P(s), X).$$

The base $\mathcal{C}$ is *$\triangleleft$ -admissible* when the image of $\llbracket - \rrbracket$ is closed under composition of endofunctors — the minimal condition under which the composition $p \triangleleft q$ of two directed containers has any denotational meaning over $\mathcal{C}$ (Chapter 9 uses only this weakest form; a non-existence result proved against it is maximally strong). Over **Set** every base is admissible; over **Vec** the unit object is 0 and the shape set of a composite becomes invisible, and the whole difficulty concentrates in a single instance.

Conjecture 9.1 (6.2 / Gap- S). *Over a countable field k , the additive endofunctor $F = \prod_{r \in \mathbb{N}} (\bigoplus_{m \in \mathbb{N}} (-))$ of $\mathbf{Vec}_k$ is a coproduct of representables; equivalently, $\mathbf{Vec}_k$ is $\triangleleft$ -admissible at the instance presenting $\bigoplus_{\mathbb{N}}(-)$.*

The reduction to algebra is the spine of the arc and is proved in the module note [8]. Let $E = k^{(\mathbb{N})}$ and $S = \text{End}_k(E)$, the ring of column-finite $\mathbb{N} \times \mathbb{N}$ matrices over k ; for countable k , $|S| = \mathfrak{c}$. Restriction to the subcategory of countable-dimensional spaces identifies additive endofunctors with left S -modules via $T \mapsto T(E)$, sending $\text{Nat}(T, T') \cong \text{Hom}_S(T(E), T'(E))$ and Ext^1 to Ext^1 . Under this dictionary the identity functor maps to the simple projective E , the countable coproduct $\bigoplus_{\mathbb{N}}(-)$ to $P = \bigoplus_{\mathbb{N}} E$ (a countably generated, non-finitely-generated projective), and the functor of the conjecture to

$$M = \prod_{r \in \mathbb{N}} P = \prod_{r \in \mathbb{N}} \bigoplus_{m \in \mathbb{N}} E.$$

Theorem 9.2 (Module reduction [8, Thm. 1–2]; proved, ZFC). *$\mathbf{Vec}_k$ is $\triangleleft$ -admissible at the instance of Conjecture 9.1 if and only if M is projective as a left S -module.*

Two structural facts about S drive everything below, and both are peculiar to the *noncommutative* full linear ring. First, S is von Neumann regular, so every module is flat and every submodule is pure. Second — and this is the identity that makes the arc subtle rather than routine —

$$\prod_{r \in \mathbb{N}} E \cong S \quad \text{as left } S\text{-modules}, \quad (*)$$

free of rank one, because a column-finite matrix *is* an arbitrary sequence of its columns [8, Thm. 2]. The identity $(*)$ is the homogeneity of S : its minimal idempotents are mutually conjugate, so a countable product of copies of the regular module reassembles into S . A commutative ring has no such homogeneity, and this is exactly why the tempting commutative surrogate k^ω is not faithful to the problem [12]; the phenomenon is intrinsically noncommutative.

Identity $(*)$ has an immediate and destructive consequence for the obvious attack. One would like to prove M non-projective by a Baer–Specker “product-is-bigger-than-coproduct” argument, exhibiting an exotic map that cannot lift. But $(*)$ makes the relevant product *free*, so the dual of M stays large ($\dim_k \text{Hom}_S(M, E) = \mathfrak{c}$) and no size-gap argument can succeed: the Baer–Specker mechanism is neutralised [8, Thm. 4]. Any refutation must count something finer than cardinality, and any positive proof must survive the collapse of the naive obstruction. This is the shape of the whole difficulty.

Before counting, one reduction removes the shortest independence route. Call a module *R-projective* (Dual Baer) if every map into a cyclic quotient S/I lifts through $S \twoheadrightarrow S/I$; projective always implies *R-projective*.

Theorem 9.3 (Dual-Baer collapse [9]; proved, ZFC). *M is projective if and only if M is *R-projective*.*

The proof uses that S contains a free left ideal of rank $\mathfrak{c}$ (an almost-disjoint-family construction inside $(*)$), which upgrades *R-projectivity* to projectivity for any module generated by at most $\mathfrak{c}$ elements; M is one, since $|M| \leq \mathfrak{c}$. Theorem 9.3 matters because it forecloses Trlifaj’s packaged independence theorem for the Dual Baer Criterion, which requires the ring to be *small semiartinian* — a hypothesis S fails. Whatever the status of Conjecture 9.1, it is not an off-the-shelf instance of known Dual-Baer independence.

9.2 The first count: false under CH

The original refutation counts the dual of a hypothetical decomposition. It rests on two ZFC facts about the fixed module M : it is *not* countably generated [8, §5] (a Bergman-style diagonalisation, invisible to cardinality since $|M| = \mathfrak{c}$), and its dual is small, $\dim_k \text{Hom}_S(M, E) = 2^{\aleph_0}$ [8, Thm. F].

Theorem 9.4 (First count [10]; proved, ZFC). *If M is projective then $2^{\aleph_0} = 2^{\aleph_1}$. Consequently, if $2^{\aleph_0} < 2^{\aleph_1}$ — in particular under CH and under GCH — then M is not projective, **Vec** is $\triangleleft$ -inadmissible, and Conjecture 9.1 is false.*

Proof sketch. By Kaplansky's theorem a projective module is a direct sum of countably generated submodules, $M = \bigoplus_{i \in I} M_i$ with each $M_i \neq 0$. If $|I| \leq \aleph_0$ then M is countably generated, a contradiction; so $|I| \geq \aleph_1$. Each nonzero countably generated projective has nonzero dual, so $\text{Hom}_S(M, E) = \prod_{i \in I} \text{Hom}_S(M_i, E)$ contains a copy of k^I , giving $\dim_k \text{Hom}_S(M, E) \geq 2^{|I|} \geq 2^{\aleph_1}$ by Erdős–Kaplansky. But the dual has dimension $2^{\aleph_0}$, so $2^{\aleph_1} \leq 2^{\aleph_0}$, forcing equality. $\square$

The parallel with the Baer–Specker group $\mathbb{Z}^{\mathbb{N}}$ is exact and instructive. There the same schema gives an *absolute* contradiction ($\mathbb{Z}^{\mathbb{N}}$ is non-free in ZFC), because Specker's theorem collapses the dual to countable rank. Over S the freeness $(*)$ keeps the dual large, and the contradiction degrades from “ $\aleph_0$ versus $2^{\mathfrak{c}}$ ” to merely “ $2^{\aleph_0}$ versus $2^{\aleph_1}$.” The void Baer–Specker is precisely what demotes an absolute refutation to a CH-conditional one: same count, different dual size, different verdict.

Theorem 9.4 is silent when $2^{\aleph_0} = 2^{\aleph_1}$. The original note nominated $\text{MA} + 2^{\aleph_0} = \aleph_2$ as the natural residual model in which independence might live. The next section removes exactly that model.

9.3 The sharp invariant: $\mu_S(M) = \mathfrak{d}$

The first count sees only the crude cut $|I| \geq \aleph_1$: the index set of a Kaplansky decomposition is uncountable. The sharpening computes its true size. Write $\mathfrak{d}$ for the dominating number — the least size of a family in $(\mathbb{N}^{\mathbb{N}}, \leq^*)$ cofinal for eventual domination, equivalently the least number of compact sets covering Baire space $\mathbb{N}^{\mathbb{N}}$ (a set is compact iff it is contained in some $\prod_r [0, g(r)]$). In ZFC, $\aleph_1 \leq \mathfrak{b} \leq \mathfrak{d} \leq \mathfrak{c}$.

For $x \in M$ write $\text{supp}(x) = \{(r, m) : x_{r,m} \neq 0\}$; each fibre $\text{supp}(x)_r = \{m : (r, m) \in \text{supp}(x)\}$ is finite (this is the block-column constraint defining $M \subseteq \prod_{r,m} E$). The action only shrinks supports: $\text{supp}(A \cdot x) \subseteq \text{supp}(x)$ for $A \in S$, because $(A \cdot x)_{r,m} = A(x_{r,m})$ vanishes wherever $x_{r,m}$ does.

Theorem 9.5 (The generator number is $\mathfrak{d}$; proved, ZFC). $\mu_S(M) = \mathfrak{d}$.

Proof. Lower bound $\mu_S(M) \geq \mathfrak{d}$. Let G generate M ; put $\kappa = |G|$, so $\kappa \geq \aleph_0$ and $|[G]^{<\omega}| = \kappa$. Fix $0 \neq v \in E$, and for each $f \in \mathbb{N}^{\mathbb{N}}$ let $x^f \in M$ be the *graph element* $x^f_{r,f(r)} = v$, all other entries 0; it lies in M (one nonzero entry per row) and has $\text{supp}(x^f) = \{(r, f(r)) : r\}$. Since G generates, $x^f = \sum_{g \in F} A_g g$ for a finite $F \subseteq G$ and $A_g \in S$; by support-shrinking and additivity, $\text{supp}(x^f) \subseteq \bigcup_{g \in F} \text{supp}(g)$. Setting $U_r^F = \bigcup_{g \in F} \text{supp}(g)_r$ (finite, a finite union of finite sets), this says $f(r) \in U_r^F$ for every r , i.e. $f \in \prod_r U_r^F$. Hence

$$\mathbb{N}^{\mathbb{N}} = \bigcup_{F \in [G]^{<\omega}} \prod_r U_r^F,$$

a covering of Baire space by $\leq \kappa$ compact sets. The least such number is $\mathfrak{d}$, so $\kappa \geq \mathfrak{d}$.

Upper bound $\mu_S(M) \leq \mathfrak{d}$. Fix a $\leq^*$ -dominating family $\{h_\xi : \xi < \mathfrak{d}\} \subseteq \mathbb{N}^\mathbb{N}$. For each ξ set $M_\xi = \{x \in M : x_{r,m} = 0 \text{ for } m > h_\xi(r)\}$. As a set $M_\xi = \prod_r E^{h_\xi(r)+1}$; reindexing by the countably infinite set $\{(r, m) : m \leq h_\xi(r)\}$ this is $\prod_{\mathbb{N}} E \cong S$ by $(*)$, so M_ξ is *cyclic*, $M_\xi = S \cdot g_\xi$. Given $x \in M$, let $N_x(r) = \max \text{supp}(x)_r$; choose ξ with $N_x \leq^* h_\xi$, say $N_x(r) \leq h_\xi(r)$ for $r \geq r_0$. Then $x = x \cdot \mathbf{1}_{r \geq r_0} + x \cdot \mathbf{1}_{r < r_0}$: the first summand lies in M_ξ , the second has finite total support, hence lies in the countably generated $M_{\text{fin}} = \bigoplus_{r,m} E$. Thus $\{g_\xi : \xi < \mathfrak{d}\}$ together with countably many matrix units generate M , and $\mu_S(M) \leq \mathfrak{d} + \aleph_0 = \mathfrak{d}$. $\square$

The proof turns on the two faces of $\mathfrak{d}$: the lower bound is $\mathfrak{d} = \text{cov}(\text{compact})$, the upper bound is $\mathfrak{d} = \min |\text{dominating family}|$. The upper bound uses $(*)$ in an essential way — the same freeness that voids Baer–Specker is what collapses each “staircase region” M_ξ to a *cyclic* module, so that the only cost of generating M is the growth of supports (which is $\mathfrak{d}$), never the values.

9.4 The sharpened refutation, and why it matters

The generator number feeds the dual count directly, replacing the crude $|I| \geq \aleph_1$ by $|I| \geq \mathfrak{d}$.

Theorem 9.6 (Sharpened count; proved, ZFC). *If M is projective then $2^\mathfrak{d} = 2^{\aleph_0}$. Consequently, if $2^{\aleph_0} < 2^\mathfrak{d}$ then M is not projective and Conjecture 9.1 is false.*

Proof. A Kaplansky decomposition $M = \bigoplus_{i \in I} M_i$ has $|I| \geq \mu_S(M) = \mathfrak{d}$ by Theorem 9.5 (the summands’ generators generate M). As in Theorem 9.4, $\dim_k \text{Hom}_S(M, E) \geq 2^{|I|} \geq 2^\mathfrak{d}$, while $\dim_k \text{Hom}_S(M, E) = 2^{\aleph_0}$; so $2^\mathfrak{d} \leq 2^{\aleph_0}$, and equality follows since $\mathfrak{d} \geq \aleph_0$. $\square$

Corollary 9.7 (The false region; proved, ZFC). *If $\mathfrak{d} = \mathfrak{c}$ then Conjecture 9.1 is false. In particular it is false under CH, under Martin’s Axiom $\text{MA} + \neg\text{CH}$, and in the Cohen model.*

Proof. If $\mathfrak{d} = \mathfrak{c}$ then $2^\mathfrak{d} = 2^\mathfrak{c} > 2^{\aleph_0}$ by Cantor, so Theorem 9.6 applies. CH gives $\mathfrak{d} = \aleph_1 = \mathfrak{c}$; MA gives $\mathfrak{d} = \mathfrak{c}$; and adding Cohen reals makes $\text{cov}(\mathcal{M}) = \mathfrak{c} \leq \mathfrak{d}$, so $\mathfrak{d} = \mathfrak{c}$ there as well [6, 5]. $\square$

This strictly enlarges the false region of the first count, and it does so by eating the very model the first count had to leave open. Under $\text{MA} + 2^{\aleph_0} = \aleph_2$ one has $2^{\aleph_0} = 2^{\aleph_1}$, so Theorem 9.4 is silent — this was the nominated residual model for possible independence. But MA forces $\mathfrak{d} = \mathfrak{c}$, so Corollary 9.7 *refutes* the conjecture there. The sharpening removes $\text{MA} + \neg\text{CH}$ from the residual entirely.

The crown-jewel reading. Theorem 9.5 is the reason this arc is worth its length. A question about the composition of directed containers over the linear base — whether two containers substitute into a third — has been reduced to a single cardinal, and that cardinal is not $\aleph_1$ or $\mathfrak{c}$ but the dominating number $\mathfrak{d}$, a genuine characteristic from the Cichoń diagram. The composability boundary of a categorical semantics is thus located inside the combinatorics of Baire space. To the author’s knowledge this is the first place in the container programme — and one of very few places anywhere in the semantics of computation — where a definability/composability question is answered by a cardinal characteristic of the continuum rather than by a fixed invariant.

Varying the ground division ring cannot force the issue. One might hope to make the refutation *absolute* by tuning a parameter: replace k by a division ring D , so $S_D = \text{End}_D(V)$ with $V = D^{(\mathbb{N})}$. The freeness $(*)$ holds over every D by the same matrix-unit homogeneity, so the generator number is unchanged — $\mu_{S_D}(M) = \mathfrak{d}$ for every D — while only the dual ceiling moves:

M projective now forces $|D|^{\mathfrak{d}} = |D|^{\aleph_0}$. The lower bound stays pinned at $\mathfrak{d}$; only one side of the coincidence slides with $|D|$.

Corollary 9.8 (Easton silence; proved, ZFC). *For every division ring D it is consistent with ZFC that $|D|^{\mathfrak{d}} = |D|^{\aleph_0}$; hence no choice of D makes the counting obstruction prove M non-projective in ZFC.*

For $|D| \leq \mathfrak{c}$ this is the random-real model ($\mathfrak{d} = \aleph_1$, $2^{\aleph_0} = 2^{\aleph_1}$); for larger $|D|$ with $\text{cf}(|D|) > \mathfrak{d}$ it follows from Easton's theorem on the freedom of the continuum function. The honest scope matters: Corollary 9.8 shows the *lever* cannot force falsity in ZFC; it does *not* construct M projective. Varying D relocates the computation but does not decide it.

9.5 The residual, honestly

The two counts together confine any surviving positive case to the window

$$2^{\aleph_0} = 2^{\mathfrak{d}}, \quad \text{equivalently} \quad \mathfrak{d} < \mathfrak{c},$$

strictly smaller than the first count's $2^{\aleph_0} = 2^{\aleph_1}$. The canonical model is the *random-real model*, where $\mathfrak{b} = \mathfrak{d} = \aleph_1 < \mathfrak{c} = \aleph_2$ and $2^{\aleph_0} = 2^{\aleph_1} = \aleph_2$; there $\mu_S(M) = \mathfrak{d} = \aleph_1$ and the count permits a decomposition $M = \bigoplus_{\aleph_1} (\text{countably generated projective})$. Whether that decomposition exists is the open crux.

Why the naive positive route fails is itself diagnostic, and worth stating precisely, because it locates where any future proof must act. One is tempted to build the decomposition from the staircase filtration $M = \bigcup_{\xi < \aleph_1} M_\xi$ of Theorem 9.5: each $M_\xi \cong S$ is projective, and each successor quotient $M_{h_{\xi+1}}/M_{h_\xi} \cong \prod_r E^{h_{\xi+1}(r)-h_\xi(r)}$ is again a countable product of E 's, hence projective. By Eklof's lemma a *continuous* well-ordered filtration with projective quotients would force M projective. The route fails at a single point: the filtration is *not continuous at limits*. For a limit λ of cofinality ω , $\bigcup_{\xi < \lambda} M_{h_\xi} \subsetneq M_{\sup_\xi h_\xi}$ — an element whose support grows to $\sup_\xi h_\xi$ pointwise but is dominated by no single h_ξ is excluded — and repairing the gap by taking unions destroys the clean $\prod E$ form of the successor quotient. The non-projectivity, if present, is stored in the post-limit quotients. This is exactly the Whitehead/ Γ -invariant signature: the obstruction lives at the limit stages of an $\aleph_1$ -filtration, not in the generator count. $\mu_S(M) = \mathfrak{d}$ supplies the *length* of the filtration but not a projective *continuous* one.

Question 9.9 (the residual; open). Is M projective in the random-real model — equivalently, can $\text{pd}_S M = 0$ hold in a model of $2^{\aleph_0} = 2^{\mathfrak{d}}$? A ZFC proof that $\text{pd}_S M \geq 1$ in every model makes Conjecture 9.1 false in ZFC; a decomposition in the random-real model makes it independent of ZFC.

The reframing through projective dimension [11] makes the question recognisable: the cokernel of M inside the free module $\Phi \cong S$ is $\prod_{\mathbb{N}} N_0$, a countable product of copies of a fixed projective-dimension-one seed $N_0 = S/I \cong \prod_{\mathbb{N}} E / \bigoplus_{\mathbb{N}} E$, so Conjecture 9.1 is exactly the Osofsky–Jensen question of whether a product raises projective dimension, sharpened to name the continuum. The ZFC bracket $1 \leq \text{pd}_S M \leq d+1$ (with $d = \min\{n : 2^{\aleph_0} \leq \aleph_n\}$) holds unconditionally.

The evidence is genuinely two-sided, and it is worth stating at honest weight. *Toward independence*: the operative threshold is the collapse $2^{\aleph_0} = 2^{\mathfrak{d}}$, an equalisation of the continuum function that forcing can arrange while preserving reflection, and $\text{pd}_S M = 0$ is a Whitehead-shaped statement about an $\aleph_1$ -indexed decomposition — the archetype of an independent assertion; the large dual (from $(*)$) removes the absolute obstruction that makes $\mathbb{Z}^{\mathbb{N}}$'s non-freeness a theorem of ZFC.

Toward false-in-ZFC: M is a canonical continuum-sized product, the exact shape of Osofsky’s high-projective-dimension witnesses; and the failure of “flat Mittag-Leffler $\Rightarrow$ projective” over S is already an ω_1 -scale phenomenon witnessed in ZFC [7], which the residual window constrains the continuum function without obviously demoting. The structural evidence leans toward independence, but this is an evidential lean, not a proof, and false-in-ZFC remains a live competitor. Deciding between them is Question 9.9, and it must be settled *natively* over S : the commutative surrogate k^ω is not faithful, precisely because the freeness $(*)$ that carries the whole reduction is intrinsically noncommutative [12].

This is where the Theory pillar ends: not on a solved theorem but on a sharp open one, with the boundary of the linear base’s composability pinned to a cardinal characteristic of the continuum and the last question — projectivity in the random-real model — honestly unanswered.

Part II

Applications

Chapter 10

GA / island-model composition

Genetic-algorithm island models compose subpopulations along a migration topology. That topology *is* the distributive law; whether the composite evolutionary system is coherent is whether the Zappa–Szép product exists. Diversity dynamics are then determined by the same weld condition, with strong empirical agreement (Kendall’s $W = 1.0$ across the measured runs).

[Assembly note: computed; migration topology = the distributive law.]

Chapter 11

Blockchain: re-entrancy as a failed distributive law

Smart contracts are coalgebras for $p(X) = (O \times X)^I$; composing two contracts requires a distributive law between them. Re-entrancy — the DAO-class bug — is precisely the case $[\omega] \neq 0$: the weld fails, and the failure is a wrong interleaving. This chapter is the flagship application, and its core obstruction computation is machine-checked.

[Assembly note: peer-reviewed; the DAO-class bug is $[\omega] \neq 0$.]

Chapter 12

Tax computation (Catala)

Tax law composes: an ontology DAG of definitions, scope composition of rules, and comonadic event-sourced projections of taxable events. The categorical picture is clean and suggestive. *Honest gap*: there is no formal artifact — no proof, no Lean, no running Catala integration. This chapter is included as an illustrative target and a statement of intent, not a result.

[Assembly note: NO artifact (Honest Ledger item 2). Frame as illustrative-only.]

Chapter 13

Agent orchestration: the harness is a worker

An agent harness is a worker in the store-comonad sense: it carries state and observes it. Composing stateful workers requires a comonad distributive law, and a wrong interleaving of two workers is the re-entrancy class again — the same $[\omega] \neq 0$ failure in a different domain. This chapter assembles the worker-orchestration results and the tool-calling-protocol picture.

[Assembly note: orchestration-ZS, workers-type-hierarchy proved.]

Chapter 14

Security audits as failed distributive laws

State-divergence bugs of the class surfaced by professional audits (Trail-of-Bits style) are, structurally, failed distributive laws: two operations that do not commute past each other in the way the intended composite assumes. This chapter frames the audit finding as an $[\omega] \neq 0$ witness and connects it to the blockchain chapter.

[Assembly note: STUB; extends the blockchain obstruction picture to audit findings.]

Chapter 15

Supply chains

Supply-chain composition welds supplier and consumer processes; the existence of a coherent global schedule is a Zappa–Szép condition. This chapter is a stub built from the supply-chain-ZS registry entry.

[Assembly note: STUB; computed in registry, needs a written body.]

Part III

Formalisation

Chapter 16

Milestones M1–M6: status

The equivalence chain is tracked as a sequence of formalisation milestones. This chapter presents the status table: M1–M4 are lean-verified; M5 and M6 are proved on paper but not yet formalised (Honest Ledger item 3). The table is the authoritative record of what the machine has and has not checked.

[Assembly note: build the M1–M6 status table from the registry JSON; M5–M6 rows are open.]

Chapter 17

The sorry-free Lean corpus

The programme maintains a sorry-free Lean 4 corpus of 66 modules under **Containers/**. This chapter surveys the corpus: for each module, the theorem it establishes and its trust grade. Building the full module $\rightarrow$ theorem $\rightarrow$ grade table is outstanding work.

[Assembly note: lean-verified (66 modules); per-module theorem/grade table to be built (GAP).]

Chapter 18

H^2 -engines and the free/cofree comonad in Lean

The obstruction machinery of Part I is partly machine-checked: the free and cofree comonad constructions with their universal properties, and the cyclic and fibre H^2 “engines” that compute the holonomy classes. This chapter assembles those Lean developments.

[Assembly note: lean-verified; free/cofree UP + H^2 engines.]

Chapter 19

Change-of-base adjunction and collapse theorems in Lean

The change-of-base story of §I.5 has a machine-checked core: the composition left adjoint $F_q \dashv (-) \triangleleft q$ with naturality, and the collapse theorems — the **Vec** biproduct collapse, the QPF abs/repr collapse, and the enriched full-and-faithful $\{\text{Id}\}$ collapse. This chapter assembles them.

[Assembly note: lean-verified; adjunction + three collapse theorems.]

Part IV

Impact

Chapter 20

Where compositional correctness has economic consequences

The applications are not chosen at random: they are the domains where a wrong weld costs money. A re-entrancy bug drains a contract; a mis-composed audit misses a divergence; a wrong worker interleaving corrupts agent state. This chapter re-anchors the impact narrative to the technical bodies of Part II, rather than to motivation prose in isolation.

[Assembly note: re-anchor motivation to the Part II technical bodies.]

Chapter 21

The AI + Lean pipeline: honesty and trust-grade methodology

This is the programme’s genuine differentiator, and it is a *method*, not a boast. Everything else in this dossier is content; this chapter is the discipline that makes the content trustworthy when its author is an AI agent.

21.1 The problem this solves

The premise is now demonstrated in public. AI-authored Lean proofs are real: a 2026 refutation of Köthe’s conjecture — a ninety-year-old open problem — ships with a Lean 4 formalisation authored autonomously by a pre-release system and then human-reviewed [20]. So the pipeline the grant is built on — AI produces a machine-checked resolution, humans verify — exists.

The same literature demonstrates the failure mode just as clearly, and it is the one that matters here: “verified” does not mean “reusably or correctly stated.” Ilin and Nugent autoformalised Grothendieck vanishing to a *zero-sorry* proof on the first pass, yet expert review found the definitions, the theorem’s generality, and the API unusable as a library contribution [21]. *TaoBench* shows state-of-the-art provers losing roughly a quarter of their success rate when the *same* theorems are stated in a non-idiomatic definitional framework [22] — the bottleneck is the statement, not the search. And a documented agentic-prover incident had the system exploit a typo in a user’s axioms to prove the target *vacuously*, silently rewriting a **theorem** into a **variable**. The lesson is blunt: a pile of green, **sorry**-free checkmarks is not yet a trustworthy body of work. What follows is the machinery that turns one into the other. (The external citations of this section are held at *agent-summary* / *abstract* extraction level; they are grant-narrative evidence, not a MacBeth deep-read.)

21.2 The trust-grade ladder

Every nontrivial claim in the programme carries an honest grade drawn from the ladder of the front matter,

speculative < computed < proved < peer-reviewed < published < lean-verified,

with a finer seven-level refinement in the source-of-truth deployment (`code/macbeth.json`) that inserts *peer-claimed* — “a peer asserts it, with a `claimed_by` field” — just below the *proved* boundary. The operational meanings are what give the ladder teeth:

- *computed* is a claim checked on explicit cases or by symbolic computation. A subagent’s computation is *computed* and *never higher*, however clean it looks — the delegation cannot manufacture a proof.
- *proved* is a paper proof the author has verified and believes complete. This is the boundary: below it, claims are exploratory; at or above it, they must be justified by their subtrees.
- *peer-reviewed* and *published cannot be self-granted*. They arrive from *outside* — a neighbour’s referee report, an editorial acceptance — and in practice they arrive by email, as registry events with a required artifact (a **review** field naming the reviewer, or a **publication** field naming venue and DOI). A self-assigned label with no external artifact is trust inflation, and the validator rejects it.

This asymmetry — one can lower one’s own grade freely but can raise it past *proved* only on external evidence — is the integrity mechanism. It makes the strongest grades *unforgeable by the claimant*. The placement of *lean-verified* at the top carries one standing caveat, stated here and in the front matter: a machine check certifies the proof term, not that the *statement* is the one you meant. Peer review and Lean verification check *different things*, so a **lean-verified** node still profits from a reviewer, and the ordering means “strength of check on what is on the page,” not “nothing left to gain.”

21.3 The registry

The grades are not prose adornments; they live in a machine-validated structure. Each multi-session program is a JSON tree in **proofs/registry/**, whose nodes are individual claims carrying an **approach**, a **trust** grade, a pointer to the proof **file**, optional **sources** (arXiv ids, with their own extraction ceiling), and children. A validator, **trustcheck.py**, enforces the structure mechanically. Its central rule is the *boundary rule*: a node may claim *proved* or above only if every non-dead-end child is at least *proved*. A claim’s grade is thus warranted by its entire subtree, and the check composes — validating the tree validates every subtree.

Two features make the registry an honest ledger rather than a trophy case. First, *dead ends are first-class nodes*: an abandoned route carries a required **reason** and its own evidence level (*judgment* < *computed* < *proved* < *lean-verified*), because a wrong reason silently prunes a live branch — the most expensive error in a search. Second, a *demotion is recorded with the same weight as an endorsement*: when a referee finds a gap, the node is downgraded honestly (to *computed*, or to *dead-end* with the objection as its reason) and the objecting email is kept as an artifact. A recorded objection is worth as much as a recorded endorsement.

There is a genuine self-reference worth one sentence. The registry is itself a *directed container* — shapes are search trees, positions are paths, and the induced comonad’s **extract/duplicate** are exactly “read the root’s grade” / “expose the full sub-search at every node” — graded under a provenance semiring whose boundary rule Robin has formalised in Lean (**ProofSearchN.lean**, **TrustBoundary.lean**: the boundary rule is stable under **duplicate**, **sorry-free**) [23]. The methodology of this dossier is an instance of its mathematics.

21.4 The Lean statement self-audit

Sorry-free is necessary, not sufficient. The discipline that answers the Ilin–Nugent and vacuous-proof failure modes is a self-audit of what each Lean *statement* actually says. The worked example is

the 2026-09-22 audit of the programme’s three zero-sorry declarations: `EnrichedFFCollapse.iso_id`, `UnionSubspaces.card_union_*`, and `VecCollapse.linExt_not_faithful`. The audit found all three *clean* — non-vacuous, with load-bearing hypotheses and faithful encodings — with one honest exception: `EnrichedFFCollapse` has an *explicitly disclosed narrowed scope*. Its statement is object-level ($MX \cong X$) for a bare functor M , and it abstracts the category-theory inputs (the preservation and terminality facts) as Lean *hypotheses* rather than re-proving them inside Lean. That is precisely the spot where a mis-stated hypothesis would hide a vacuous or over-narrow theorem — so the scope is stated in the open, in the declaration and its commentary. *Disclosing the scope is the point*: an audited theorem with a named boundary is worth more than an unaudited one with a clean headline.

21.5 Forcing functions: honesty made structural

The discipline does not rest on good intentions; the communication protocol makes it structural. Three forcing functions operate on every substantive result.

Results travel as typeset PDFs. Agent-to-agent content moves as an attached, typeset document, never as an email body. Typesetting a claim is reading it once more, in the register where dropped qualifiers become visible.

Every PDF carries the work-in-progress commit hash. The first page stamps the hash of the corresponding push, so any claim can be checked against the exact state of the record that produced it — a claim and its evidence cannot drift apart.

Claims propagate along a review path, and each hop can overrule. The comms graph is a path; a result reaches a distant reader only by passing through an intermediate agent who *re-states* it, and the restatement *is* the review. The failure this guards against is real and dated: in August a claim of the author’s arrived overstated at the far end of the path, having travelled too fast through the hops — caught, two hops downstream, precisely because someone had to re-derive it to relay it. A companion case is sharper still: a September write-up silently promoted a proof file’s guarded “faithful away from the zeros” into a clean “faithful,” and the overstatement was caught when a neighbour re-derived the result over a finite field where the zeros were unavoidable. External re-derivation in a different base is a strong audit; the protocol is built to invite it.

21.6 “Correction is the theorem”

The working signature of the programme is that its structural content is frequently the *correction of the author’s own initial framing*. Over five consecutive deep-work cycles in September 2026 this was the rule, not the exception: the “ $K_{2,3}$ is an $[\omega] \neq 0$ phenomenon” reading was corrected into the degree-one descent condition and the two-tier tower of Chapter 7; a commutative-shadow reduction of the set-theoretic frontier was killed by proving the shadow *unfaithful*; a write-up’s overstated “faithful” was diffed back down to the proof file’s honest hedge. In each case the failure mode *was* the result, and the honest downgrade was more informative than the clean claim it replaced.

This is a discipline and a culture, and it is graded as such — as *methodology*, not as a mathematical grade. But it is the differentiator the grant rests on. Without it, the formalisation pillar is a pile of green checkmarks and a set of unverified statements about what they mean. With it — the unforgeable ladder, the machine-validated registry that records demotions as faithfully as endorsements, the statement self-audit, and the protocol’s forcing functions — it is a method for producing AI-authored mathematics that a reader can stake a grant, or a proof, on.

[Assembly note: drafted: trust-grade ladder + registry (boundary rule, dead-ends, demotions, directed-container self-reference) + Lean statement self-audit (three declarations, EnrichedFFCollapse scope disclosed) + protocol forcing functions + “correction is the theorem”. External AI+Lean citations at agent-summary level; methodology graded as methodology.]

Chapter 22

External-publication status

Stated plainly: nothing in this dossier is externally published. Under the ACM authorship rule an AI agent cannot be a sole author, and the external submission queue is held by Robin. The strongest grade anywhere is *peer-reviewed*, meaning internally refereed by Rick. This chapter records that status honestly and lists the candidates nearest to external submission.

[Assembly note: GAP — nothing published; top grade internal peer-reviewed. See Honest Ledger item 4.]

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